

# How to Tackle Threats to Cloud Security

A large number of SMEs have found out that cloud computing has many advantages, which has led to a surge in its adoption. That's precisely why security concerns in the cloud need to be addressed urgently. This article is a primer on cloud security.



**T**his article covers some of the top security threats on the cloud, what their implications are and some measures to overcome them or reduce their impact.

## Cloud computing

Cloud computing involves the use of a large number of servers to allow users data storage, data access and data manipulation, along with the provision for other services. As Wikipedia aptly defines it, "Cloud computing is Internet-based computing, in which large groups of remote servers are networked to allow centralised data storage and online access to computer services or resources."

Cloud computing has different service models that include the Software-as-a-Service (SaaS), Platform-as-a-Service (PaaS) and the Infrastructure-as-a-Service (IaaS) options. Cloud computing is also categorised into public, private and hybrid clouds.

## Security in the cloud

'Security in the cloud' implies security of the data stored and the services that are offered to a particular organisation via the cloud. It includes the various policies, technologies and controls that are used to protect data, applications and the associated infrastructure within the cloud. There are several security issues associated with cloud computing. These are mainly classified into those faced by cloud service providers and the ones faced by their customers.

## Top threats to cloud security

### Data breaches

A data breach is when important and sensitive data gets leaked. Every chief information officer (CIO) takes precautions to ensure that no breach of data occurs; else, internal data may fall into the hands of competitors or attackers. If there is some flaw in the design of the cloud services database, then attackers can access sensitive information from all the client applications, thus posing a serious threat to the data on the cloud.

However, data can be encrypted to reduce the impact of the breach so that even if there is a leakage, the enemy cannot decipher the encrypted information. The best practice here is to use split key management and homo-morphic key encryption. The first option splits your encryption key into parts. One part is handled by the cloud security application and one part is always handled only by you. Both parts are required to decrypt your data. This way, even if one part is used illicitly, your data cannot be breached. The second measure, homo-morphic key encryption, is the only way to ensure that the key itself is encrypted even while in use.

Various policies and procedures should be established to prevent leakages, especially when data is being shared. There should also be various policies and assessment procedures for privacy enhancement to ensure enough privacy as required by